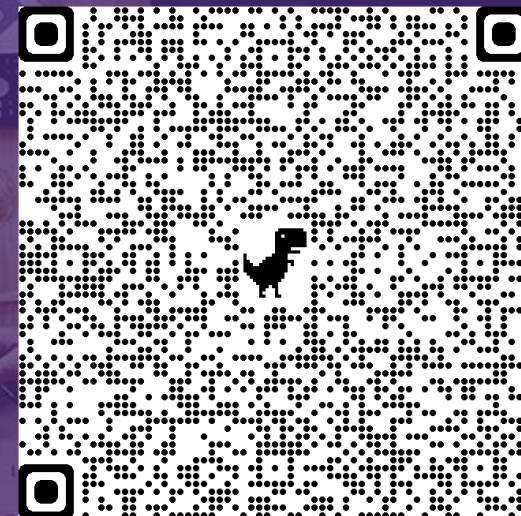


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Puppet Debugger

Campinas, 06Mai22



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Puppet Debugger

About myself



Roberto Nogueira

Software/Telecommunications Engineer

São José dos Campos, São Paulo, Brazil · 500+ connections

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Universidade de São Paulo



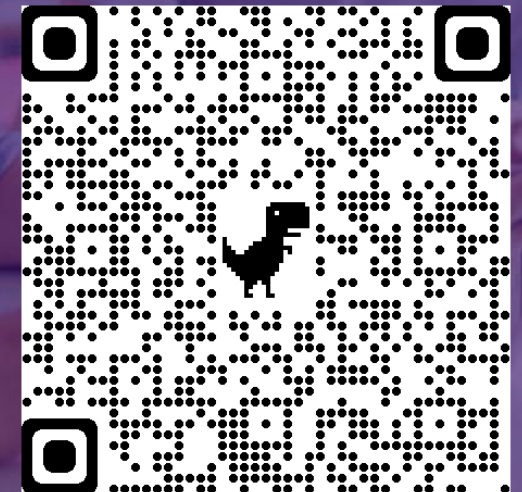
Blog

Systems Specialist at Daitan (now Encora) since 2021. Graduated in Software Engineering and Telecommunications, with more than 25 years of experience in the area and Experienced Integrator certified by **Ericsson**

Email : roberto.nogueira@encora.com

LinkedIn: https://www.linkedin.com/public-profile/settings?trk=d_flagship3_profile_self_view_public_profile

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Puppet Debugger

Contents

1. Objective
2. Puppet
3. Puppet Language Basics
4. Puppet Debugger
5. Puppet Debugger Commands
6. Puppet Extensions
7. References

Puppet Debugger

Objective

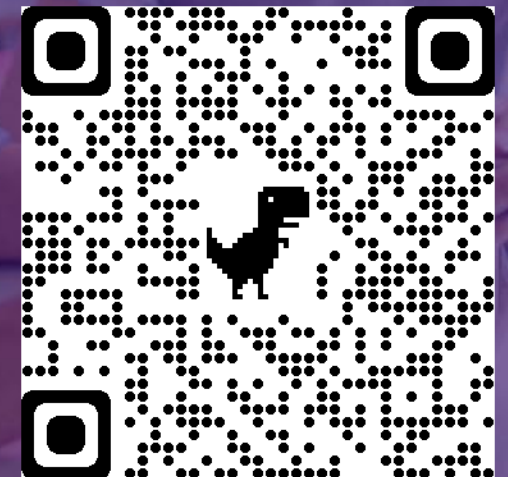
It is to introduce a [Puppet Debugger](#), a command line tool to help you write, understand, and debug Puppet code. In order to achieve that the Puppet and its language is revised.

PuppetDebugger



A command line tool to help you write, understand, and debug puppet code.

[Demo](#) is an Online Puppet Debugger.



“

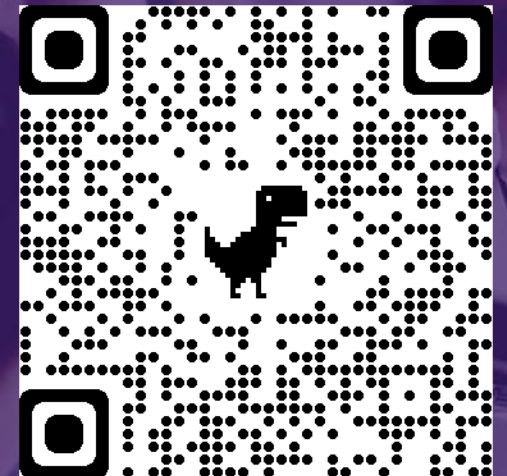
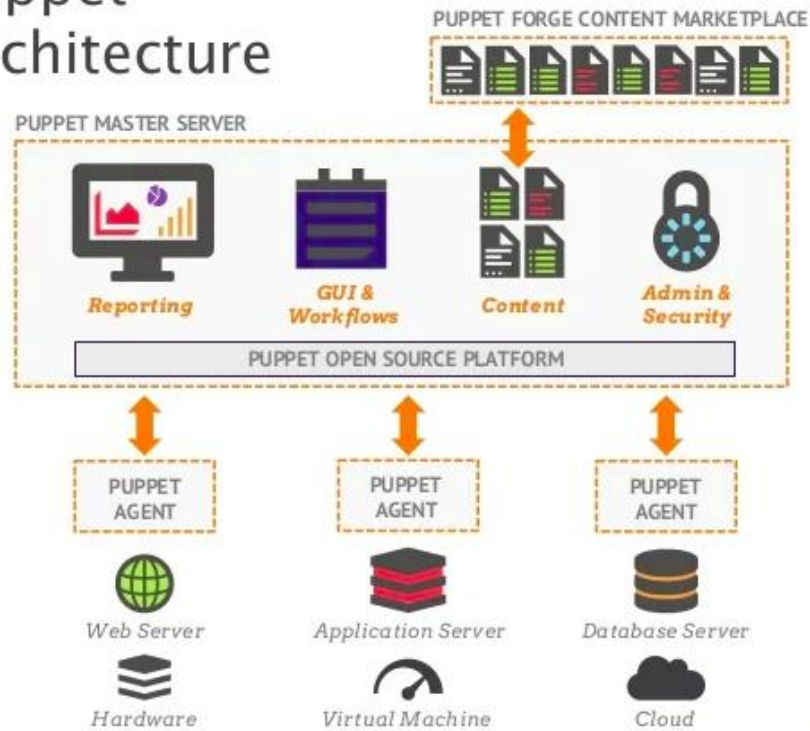
A command line tool to help you
write, understand, and debug
Puppet code

Puppet

Architecture

Puppet is a software configuration management tool which includes its own declarative language to describe system configuration.

Puppet Architecture



Puppet

Installation

Install Puppet and Puppet Debugger

```
$ docker run --name=debian11 -t -i --rm debian:11 bash  
  
$ update  
$ apt install systemctl wget lsb-release -y  
$ wget https://apt.puppetlabs.com/puppet7-release-bionic.deb  
$ dpkg -i puppet7-release-bionic.deb  
$ apt update  
$ apt-get install puppet-agent -y  
$ echo 'export PATH=$PATH:/opt/puppetlabs/puppet/bin' >> ~/.bashrc  
$ source ~/.bashrc  
$ gem install --no-document puppet-debugger
```

Puppet relevant commands

```
# print puppet configuration, usually has to be execute with sudo  
$ puppet config print  
:  
# set log level to debug, usually has to be execute with sudo  
$ puppet config set log_level debug  
:
```

In order to install **Puppet** the way it is described here [Docker](#) is required see [Ref.1](#) for further installation.

Puppet Debugger

Puppet Language Basics

- Introduction
- Resources
- Classes
- Modules
- Variables
- Metaparameters

Puppet Language Basics

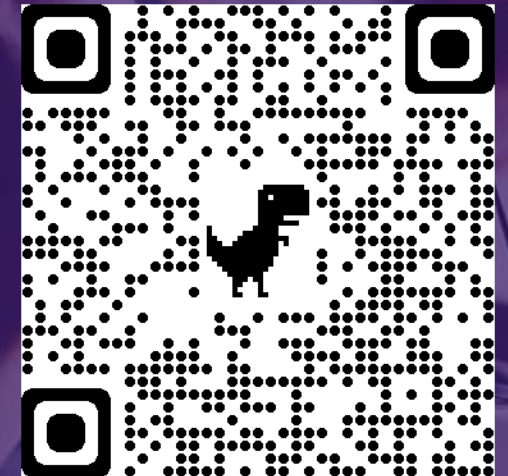
Introduction

Puppet consists of a custom [declarative language](#) to describe system configuration, which can be either applied directly on the system, or compiled into a catalog and distributed to the target system via [client-server paradigm](#) (using a [REST API](#)), and the agent uses system specific **providers** to enforce the resource specified in the manifests. The resource abstraction layer enables administrators to describe the configuration in high-level terms, such as users, services and packages without the need to specify OS specific commands (such as [rpm](#), [yum](#), [apt](#)).

DESCRIPTION



IMPLEMENTATION



Puppet Language Basics

Resources

Are the fundamental **building blocks** of the Puppet Language.

Is characteristic of a **node** that needs to be managed overtime.



Resources Types are e.g. User, Package and Service.

Puppet Language Basics

Resources

Resource Declaration adds a resource to the catalog and tells Puppet to manage that resource state.

```
resource_type { 'name':  
  attribute => value,  
}
```

```
file { '/etc/motd':  
  ensure   => 'file',  
  content  => 'Hello World',  
  owner    => 'root',  
  group    => 'root',  
  mode     => '0644',  
}
```

Puppet Language Basics

Resources Examples in Debian

Check the existent resources types

```
$ puppet resource --typesadd
:
file
package
:
```

Check the existent resources **file** type

```
$ puppet resource file '/etc/hosts'
:
file { '/etc/hosts':
  ensure => 'file',
  content =>
    '{sha256}674a2d0859a6ed81ce1e3513f6ee1f710b47c3b97301b284124d5b95cf566
    b47',
  ctime  => '2022-05-21 12:17:40 +0000',
  group  => 0,
  mode   => '0644',
  mtime  => '2022-05-21 12:17:40 +0000',
  owner  => 0,
  provider => 'posix',
  type   => 'file',
}
```

Puppet Language Basics

Resources Examples in Debian

Check the existent resource **service** type

```
$ service --status-all
[ ? ] hwclock.sh
[ - ] procps

$ puppet resource service 'procps'
service { 'procps':
  ensure => 'running',
  enable => 'false',
  provider => 'debian',
}
```

Installing a resources **package** type

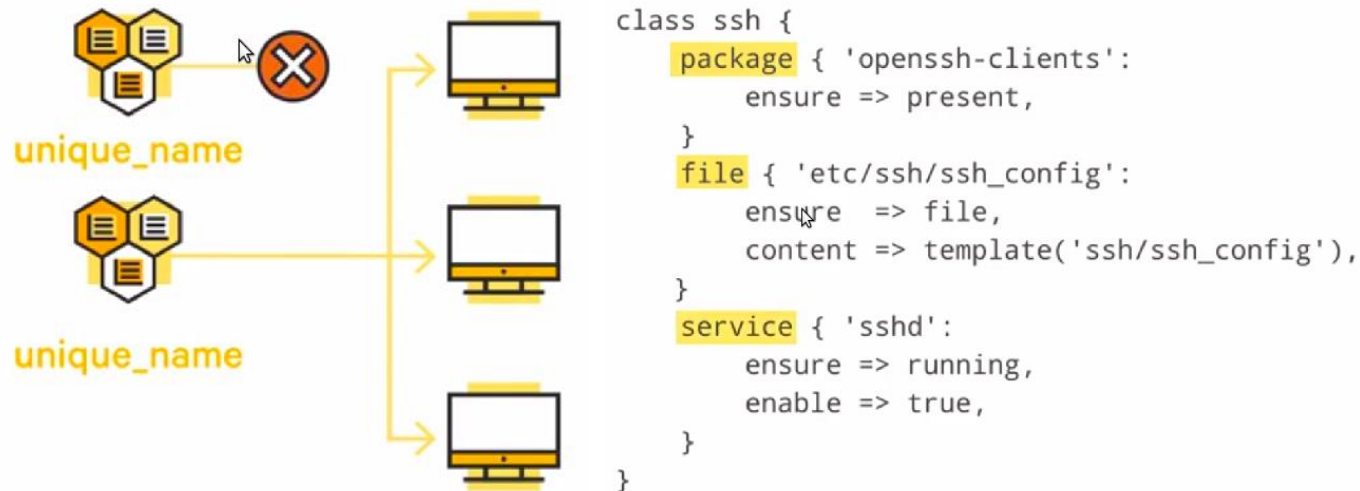
```
$ puppet resource package 'tree' ensure=present
Notice: /Package[tree]/ensure: created
package { 'tree':
  ensure => 'purged',
  provider => 'apt',
}
```

See also in [Resources](#) of [Puppet Language Basics](#) in Puppet channel in YouTube and homepage.

Puppet Language Basics

Classes

Multiple Puppet Resources can be collected and distributed together using **Classes**, by combining Resources into a Class. A Class can configure a [large block of functionality](#).

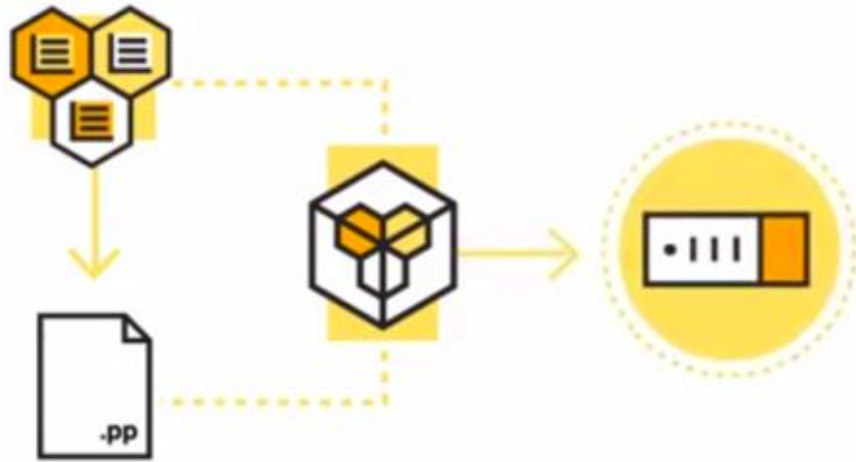


See also in [Classes](#) of [Puppet Language Basics](#) in Puppet channel in YouTube and homepage.

Puppet Language Basics

Modules

Modules are one of the most basic building blocks of the Puppet Language. They are used to manage a specific task or an application such as installing or configuring a piece of software.



Puppet Language Basics

Modules

Module is simply a directory tree that contains the files needed for a given configuration. This directory structure allows nodes to discover and loads Classes, Resources Types and more.

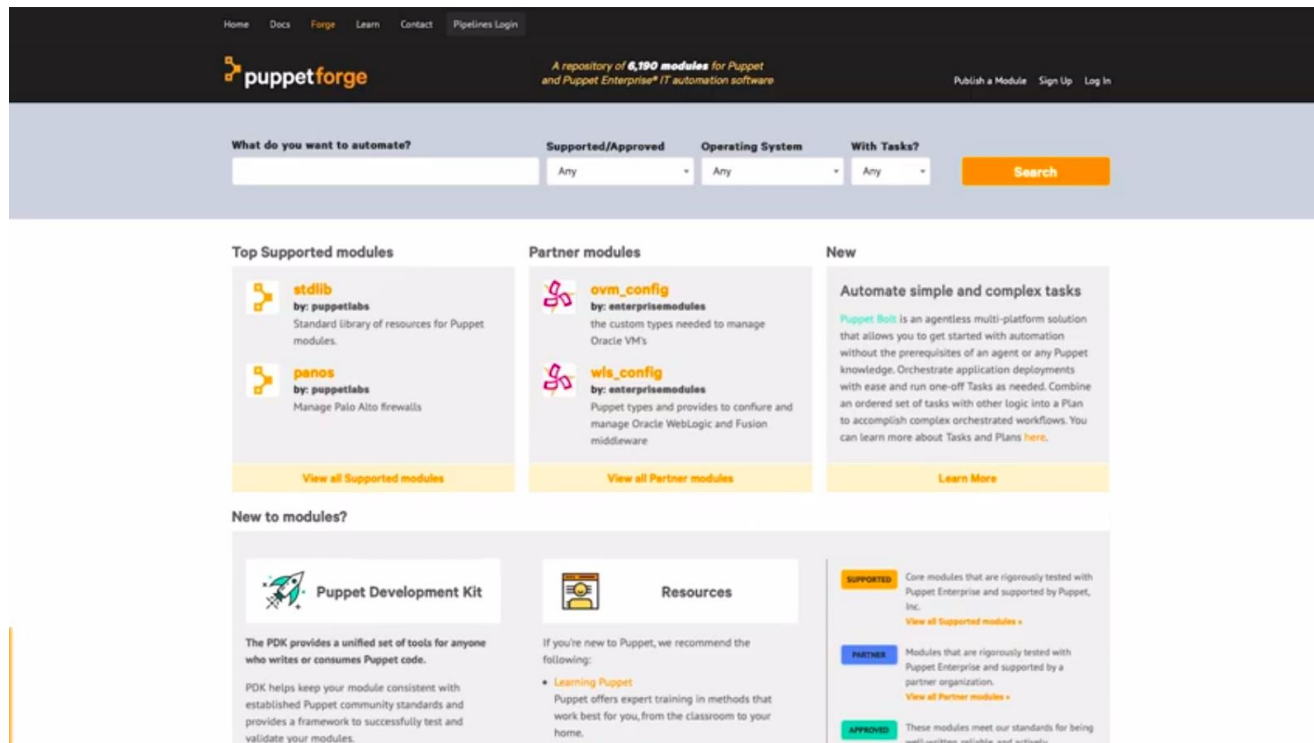
```
module-name/  
|- manifests  
    |- init.pp  
|- files  
|- templates  
|- examples  
|- spec  
|- metadata.json
```

module-name is the name of the top's directory tree. If a Class is defined in a Module you can declare that name in any Manifest. When Puppet Master compiles the catalog it will dynamic load the Manifest that contain the Class definition.

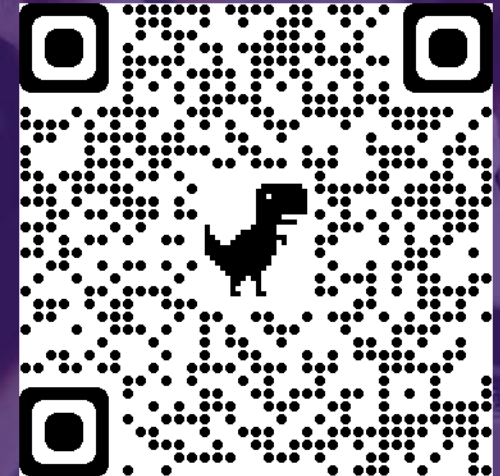
Puppet Language Basics

Modules

There are thousand of Modules available for downloading at [Puppet Forge](#), written by Puppet developers and community members. There are compatible with multi-platforms.



The screenshot shows the Puppet Forge website interface. At the top, there's a navigation bar with links: Home, Docs, Forge, Learn, Contact, Pipelines, and Login. Below this is a search bar with filters for 'What do you want to automate?', 'Supported/Approved', 'Operating System', and 'With Tasks?'. The main content area is divided into three columns: 'Top Supported modules' (listing stdlib and panos), 'Partner modules' (listing ovm_config and wis_config), and 'New' (featuring Puppet Bolt). Below these is a 'New to modules?' section with links to the Puppet Development Kit, Resources, and a list of module categories (Supported, Partner, Approved).



Puppet Language Basics

Modules

Puppet Module relevant commands

```
# list puppet modules and its path, usually has to be execute with sudo
$ puppet module list
:
$ puppet config print modulepath
:
# install puppet modules, usually has to be execute with sudo
$ puppet module install stdlib
:
```

See also in [Modules](#) of [Puppet Language Basics](#) in Puppet channel in YouTube and homepage.

Puppet Language Basics

Variables

Variables exists in order to be interpolated in strings to avoid rework in case the value is updated.

```
file { 'path/to/file/share/directory1': $file_share_path = 'new/path/to/file/share'
  ensure => directory,
}
file { 'path/to/file/share/directory2': }
file { 'path/to/file/share/directory3': }

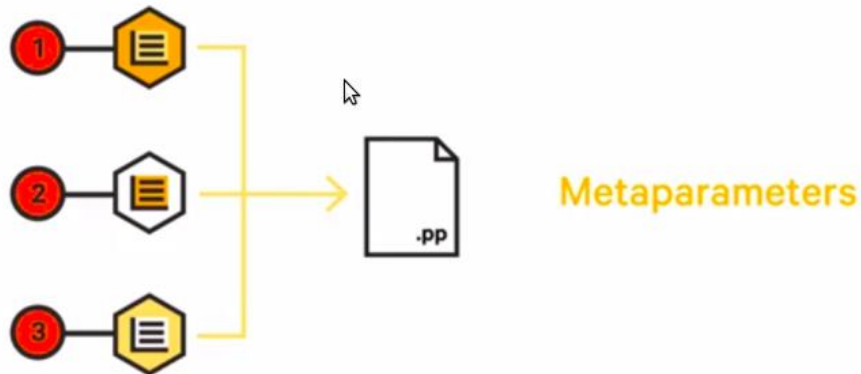
file { "${file_share_path}/directory1":
  ensure => directory,
}
file { "${file_share_path}/directory2":
  ensure => directory,
}
file { "${file_share_path}/directory3":
  ensure => directory,
}
```

See also in [Variables](#) of [Puppet Language Basics](#) in Puppet channel in YouTube and homepage.

Puppet Language Basics

Metaparameters

When a Puppet manifest is loaded the resources inside the manifest are managed the order they are declared. But when you need a group of resources to be managed in different or specific order you must explicit declare the relationships there where the **Metaparameters** come in. **Metaparameters** are special attributes that can be used in any resource type, and specify how your system act to the resource rather than mapping directly to the system state.



Puppet Language Basics

Metaparameters

The Puppet Language uses four Metaparameters to establish relationships.

- **'before'** causes a resource to be applied before the target resource.
- **'require'** causes a resource to be applied after the target resource.
- **'notify'** causes a resource to be applied *before* the target resource, but will refresh only if the notifying resource changes.
- **'subscribe'** causes a resource to be applied *after* the target resource, again refreshing only if the target resource changes.

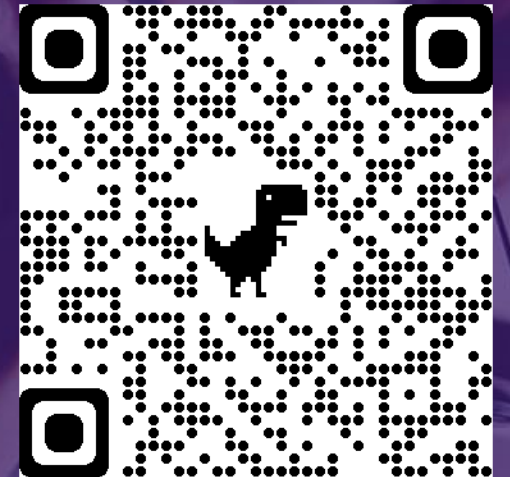
```
file { '/etc/ssh/sshd_config':  
  ensure => file,  
  mode   => 600,  
  source => 'puppet:///modules/sshd/sshd_config',  
  require => Package['openssh-server'],  
}  
  
package { 'openssh-server':  
  ensure => present,  
  before => File['/etc/ssh/sshd_config'],  
}  
  
service { 'sshd':  
  ensure => running,  
  enable => true,  
  subscribe => File['/etc/ssh/sshd_config'],  
}
```

See also in [Metaparameters](#) of [Puppet Language Basics](#) in Puppet channel in YouTube and homepage.

Puppet Debugger

Features

- Puppet Code and Language Evaluator
- Fully Featured **REPL**
- Mock real or fake systems using **facterdb**
- Works with puppet **4.x**, **5.x**, and **6.x**
- Easy Gem installation
- Easy Code Playback
- Set Breakpoints
- List all functions, datatypes, types
- Plugin system for additional functionality
- Test snippets of code in realtime



Puppet Debugger

Puppet Debugger Commands

The debugger comes with many commands that allow you to inspect the catalog, your code or reference items currently in scope. All commands are plugins that can process, modify or perform some action on the input and/or output of the debugger.

```
1:>> commands
Context
  environment    Show the current environment name
  reset          Reset the debugger to a clean state.
  whereami       Show code surrounding the current context.

Editing
  play           Playback a file or URL as input.

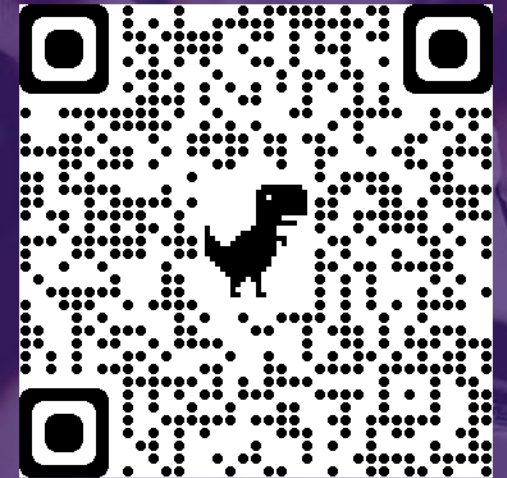
Environment
  datatypes      List all the datatypes available in the environment.
  functions       List all the functions available in the environment.
  types          List all the types available in the environment.

Help
  commands       List all available commands, aka. this screen
  exit           Quit Puppet Debugger.
  help           Show the help screen with version information.

Node
  classification  Show the classification details of the node.
  facterdb_filter Set the facterdb filter
  facts          List all the facts associated with the node.

Scope
  classes        List all the classes current in the catalog.
  krt             List all the known resource types.
  resources       List all the resources current in the catalog.
  set             Set the a puppet debugger config
  vars           List all the variables in the current scopes.

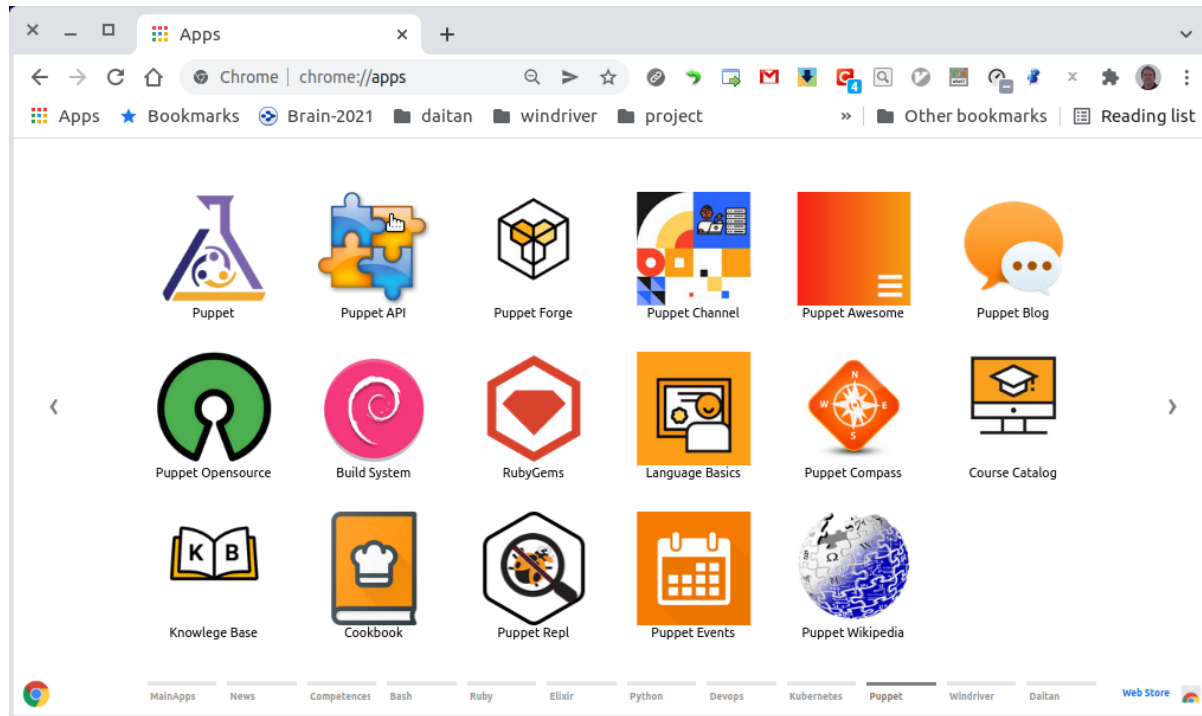
Tools
  benchmark       Benchmark your Puppet code.
```



Puppet Debugger

Puppet Extensions

[Chrome](#) and [VSCode](#) can help you stay productive and get more out of your browser and code editor.

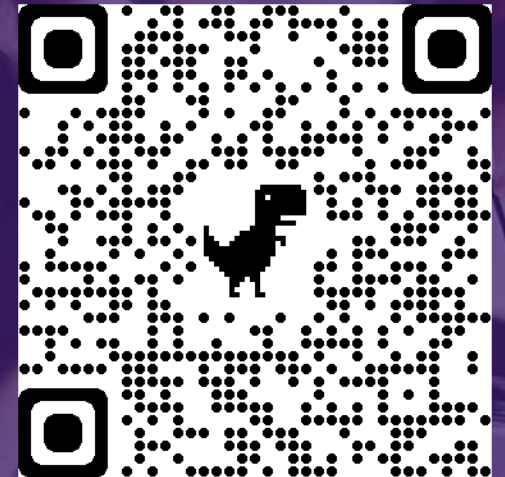


[Puppet](#) - Chrome apps for Puppet configuration management.

For further info about Chrome Apps check [Create and publish custom Chrome apps & extensions](#).

[puppet.puppet-vscode](#) - Puppet Visual Studio Code Extension.

For further extensions check [VSCode marketplace](#).



Puppet Debugger

References

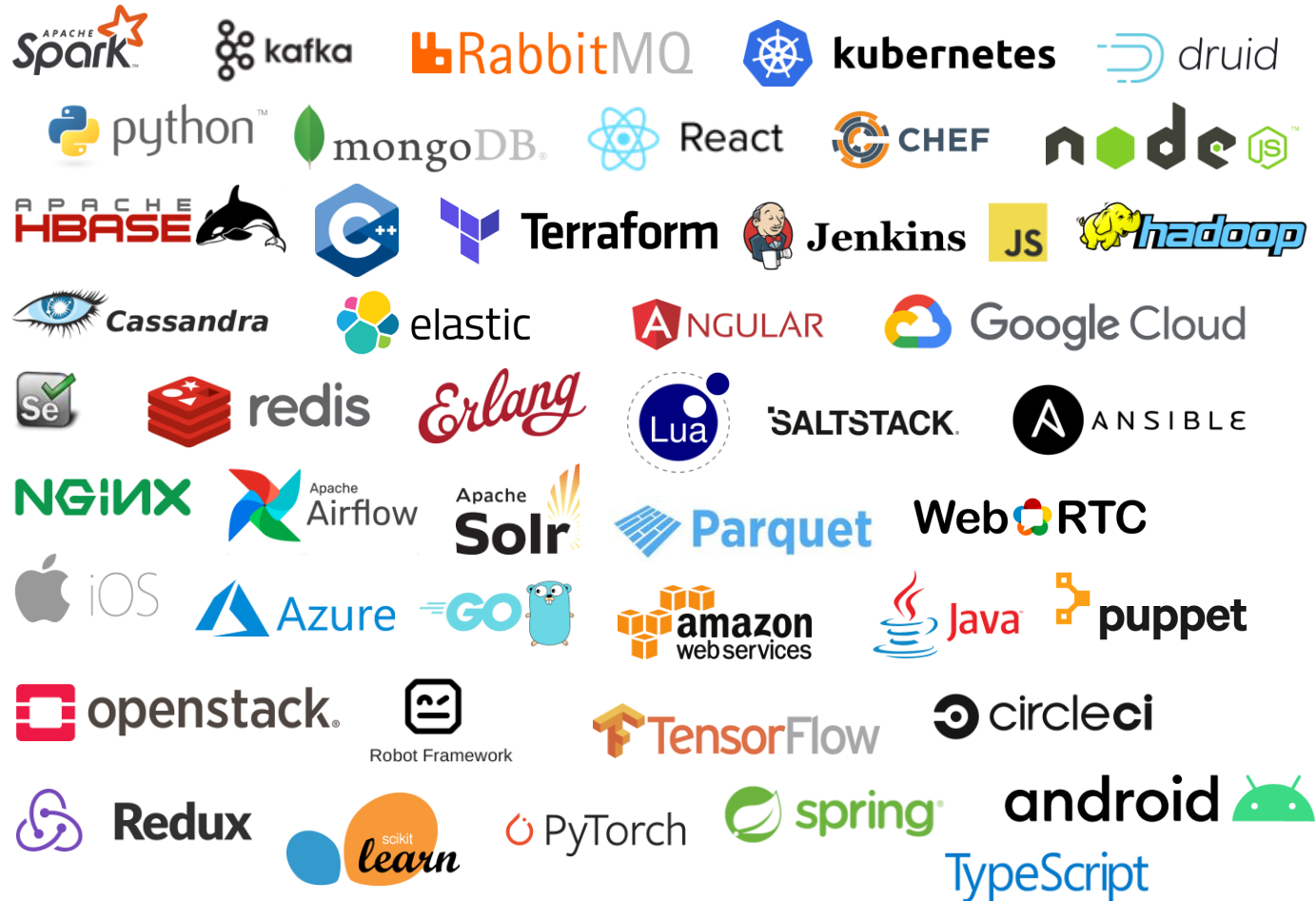


1. [Puppet in Wikipedia](#)
2. [Docker Installation](#)
3. [Puppet Language Basics](#)
4. [Puppet Forge](#)
5. [Puppet Debugger](#)
6. [Using the Puppet Debugger for Lightweight Exploration - Corey Osman](#)
7. [Puppet VS Code Tips and tricks](#)
8. [Puppet YouTube Channel](#)
9. [Puppet Cookbook](#)
0. [Awesome Puppet](#)

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